

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in-between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Gathered Materials	Hammer and Drill
Step 2	Put together the foundation	Drill, Screwdriver, Hammer, Jigsaw
Step 3	Built the spot for the plant tray to go	Same tools as step 2
Step 4	Built the arms for the shade	Drill, Hammer
Step 5	Made the rotator for the roof	Drill, cylinder of wood, nails, hammer
Step 6	Made the actual shade part	Privacy Shield, spray paint
Step 7	Connected the battery and parts	String, Battery, Motor, Wires

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Date:

Photos with captions of your full construction process from day 1 of building through to the

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
2x4	Wood	Not sure	Not sure -	2	Not sure
1x4	Wood	Not sure	Not sure	8	Not sure
Cylinder of Wood	Wood	Not sure	Not sure	1	Not sure
Shade	Privacy Shield	Not Sure	Not sure	1	Not sure
Motor	Makes the string be pulled up/down	Website Somewhere	I don't know	1	Not sure
Battery	Powers the motor	Off of a website	I don't have a clue	2	Not sure
Wires	Connect the power from the motor to battery	Hardware store	99 cents	4	5

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Must be programmable	Allow it to be hands free and be at times
Must lift the roof	Can have the strength do that
Must hold 10lbs	Have a strong, stable support
Has to be automatic	No human touch must happen for it to work

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	Description	Describe your prototype's functionality to test this attribute
Folding Roof	Allows sunlight to go on and off of the plant	Need a motor and pulley system
Stable Support	Allows the tray to hold 10 lbs and to stay stable	Must be able to hold 10lbs
Pulley/String System	Allows the Roof to go up and down depending on the amount of sunlight	Has to be able to lift the roof
Motor	Allows the string to be pulled which in turn makes the roof go up	Has to have the strength to pull the roof